Cameron MacGillivary

226-927-5048 | c.macgillivary@outlook.com | linkedin.com/in/cameron | github.com/cameron

Education

University of Waterloo — Waterloo, ON
Honours Bachelor of Computer Science — *Sep. 2026 – Present*

Catholic Central High School — London, ON
International Baccalaureate Certificate Programme & Ontario Secondary School Diploma — *Sep. 2022 – Jun. 2026*

Technical Skills

Languages: C++, Scheme (Racket), Bash, LaTex
Developer Tools: Linux, Neovim, Git, GitHub, Ninja, CMake, Visual Studio, Excel, Tailscale, Kanata, systemd
Programming Concepts: Object Oriented Programming, Data Structures (lists, vectors, binary search trees)

Experience

Agriculture Equipment Service Technician — May 2023 – Aug. 2026
Dortmans Bros Farm Equipment — *Strathroy, ON*

- Outfitted 5+ brand new facilities with Genius, Meridian, Vereijken, and ValMetal equipment and infrastructure.
- Serviced, maintained, and installed feed, water, ventilation, and other critical equipment in livestock facilities.
- Performed troubleshooting and repairs to ensure reliable operation and the health of the livestock.

Vex V5 Robotics Team Lead — Oct 2023 – Jun. 2026
CCH Robotics Team — *London, ON*

- Oversaw logistics for the team (20+ people) and coordinated preparation for robotics competitions.
- Organized team activities, delegated responsibilities, and supported competition readiness.
- Collaborated with team members on robot development, assembly, and competition preparation.

Cub/Venture/Rover Scout — Sep. 2016 – Present
1st Strathroy Scouting, Scouts Canada — *Strathroy, ON*

- Organized and participated in camping trips to local and international destinations.
- Planned and led meetings to teach youth wilderness survival skills.
- Supported logistics and event operations at a local festival.

4-H Dairy Club Member/Vice President — May 2018 – Nov. 2022
Forest 4-H Dairy Club — *Forest, ON*

- Directed club meetings and assisted new members with their 4-H projects.
- Participated in local, provincial, and national livestock shows.
- Trained and cared for dairy calves, including feeding, watering, washing, and cleaning pens.

Projects

Homelab | *C++, Linux, TCP Sockets, Tailscale, Git, systemd* — July 2026 – Present

- Developed a C++ client-server application using TCP sockets to communicate with a Linux server.
- Designed a modular project architecture separating client, server, and shared components using CMake.
- Configured and deployed a systemd service to run the server as a restricted, non-privileged user.
- Managed a self-hosted minimal Arch Linux environment.

Custom Keyboard and Input Configuration | *Linux, systemd, Kanata, ZMK* — July 2025 – June 2026

- Configured Kanata using Lisp-like configuration files to implement custom keybindings across different keyboards.
- Created a systemd service to start the remapping utility at boot as a background daemon.
- Built a custom wireless split mechanical keyboard with LED displays and rotary encoders using ZMK firmware.
- Created custom keyboard layouts with multiple layers, home-row modifiers, and key macros.